

A Study on Multisensory English Pronunciation Learning Support Based on Speech Prosody Visualization

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With the development of mobile applications and online platforms, pronunciation training based on digital tools has been widely adopted in English learning. However, many existing language learning systems primarily rely on synthesized or relatively mechanical example audio, whose intonation variation is often limited. As a result, learners may imitate pronunciations that lack natural prosody. In addition, some learning approaches focus mainly on word-level practice. When learners place individual words into complete sentences, they often find it difficult to master the correct intonation and stress patterns, which in turn affects the naturalness and fluency of their speech. To address these issues, this study proposes a pronunciation learning support method based on speech prosody visualization. Acoustic features such as pitch and intensity are extracted for each word in a sentence through speech analysis. These features are then mapped to visual variables, including text size, color, and vertical position, forming multiple visualization representations. Different visual mapping schemes are constructed, and user testing is conducted to evaluate the effectiveness of these designs in representing speech prosodic information and assisting sentence reading. This study aims to explore how visualization methods can help non-native English learners more intuitively understand sentence-level speech prosody, thereby improving their perception of intonation patterns and the effectiveness of pronunciation practice.

CCS CONCEPTS • Applied computing → Computer-assisted instruction • Human-centered computing → Empirical studies in HCI

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1 INTRODUCTION

Speech prosody is an important component of spoken English expression, mainly manifested in variations of pitch, speech intensity, and stress distribution. Accurate prosody not only affects the naturalness of language expression but also directly influences listeners' understanding of meaning and emotion. However, for non-native English learners, relying solely on auditory perception often makes it difficult to accurately detect intonation changes and stress positions within a sentence, they have the difficulty in the learning process, because they found it is hard to detect and locate the

mispronounced parts in their own speech while practicing [2]. Especially in natural speech flow, where acoustic features vary in complex ways, learners often find it difficult to quickly identify and locate the parts of pronunciation that need adjustment.

At the same time, many existing English learning applications provide example audio primarily in the form of synthesized speech. Such audio typically exhibits relatively flat intonation patterns and lacks the natural prosodic variation present in authentic speech, which may lead learners to develop overly mechanical pronunciation when imitating these examples [4]. In addition, practicing with audio examples often requires repeated listening and comparison, which increases the learning effort and does not facilitate quick memorization or understanding of prosodic patterns.

On the other hand, some learners tend to practice pronunciation at the level of isolated words [5]. When these words are placed into complete sentences, learners often struggle to determine the appropriate sentence-level intonation structure and stress placement, resulting in unnatural reading or difficulty recalling the correct pronunciation patterns.

2 METHODOLOGY

According to Mayer’s Cognitive Theory of Multimedia Learning (CTML) [1], integrating verbal and visual information can significantly enhance learners’ understanding and facilitate knowledge acquisition. Therefore, this study extracts acoustic features such as pitch and intensity from speech signals and maps them.

2.1 Speech Analysis

The OpenAI Whisper speech recognition model is used to transcribe the .wav audio file and obtain timestamp information for each word in the sentence. Based on the timestamp segmentation, the Librosa audio analysis library (Librosa 0.11.0) is used to extract acoustic features from each word segment. The average values of pitch (Hz) and intensity (dB-log energy) are calculated for each word [3]. Table 1 shows the start and end times of each word in the .wav audio sentence, as well as the average pitch and average intensity for each word.

Table 1: Extraction of acoustic features from .wav audio Files

Word	Starting Time Node	Temporal End Node	Avg_pitch	Avg_intensity
Hey,	0	0.86	113.851	-134.701
Jack,	1.36	1.66	120.477	-15.72
it's	2.14	2.42	/	-24.923
time	2.42	2.64	137.257	-21.003
to	2.64	2.86	121.351	-20.002
wake	2.86	3.08	95.013	-17.355
up	3.08	3.36	107.694	-23.072
and	3.36	3.52	97.366	-27.305
get	3.52	3.68	99.112	-21.294
out	3.68	3.86	95.607	-16.999
of	3.86	3.96	84.849	-23.521
bed.	3.96	4.2	88.913	-22.815

2.2 Data Processing

According to the overall numerical range of the extracted pitch and intensity data, the values are divided into three level intervals: low, medium, and high. These levels are used for subsequent visual variable mapping in order to enhance the visual distinction between different levels of speech features [2].

2.3 Visualization Design

The acoustic features after range classification are mapped to different visual variables, including text size, color, and vertical position. A total of six single-variable visualization schemes are designed. Figure 1 shows the vertical position visualization of pitch and intensity features for each word in the utterance. The vertical axes are linearly scaled to the range of feature values (pitch in Hz; intensity in dB log energy). Each word is mapped to a vertical position proportional to its average feature value. Figure 2 shows the color visualization of pitch and intensity features for each word in the utterance. Each word is assigned one of three colors according to its feature level (high/medium/low), as defined by the threshold ranges, to visually represent intonation variations [7]. Figure 3 shows the Text size visualization of pitch and intensity features for each word in the utterance. Words are assigned one of three font sizes according to their feature levels (high/medium/low), defined by equal-height threshold ranges, to show relative prosodic prominence.

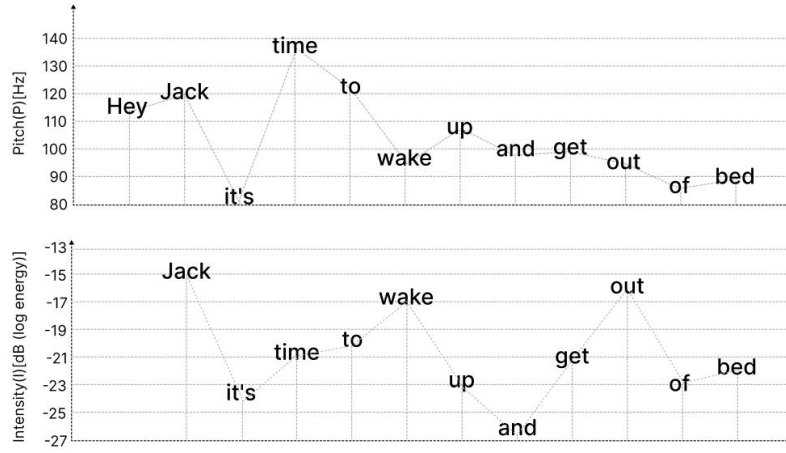


Figure 1: Vertical position visualization of pitch and intensity features for each word in the utterance.



Figure 2: Color visualization of pitch and intensity features for each word in the utterance.

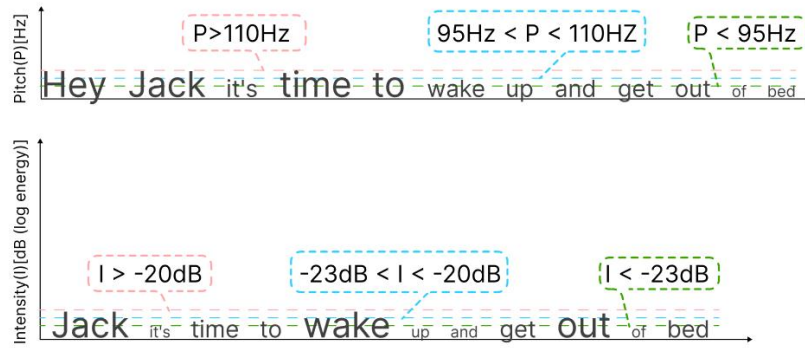


Figure 3: Text size visualization of pitch and intensity features for each word in the utterance.

2.4 User Evaluation

Finally, in the user evaluation stage, a user study is conducted through an online questionnaire. Participants first listen to the original speech sentence and then observe different visualization representations. They are asked to evaluate each

design according to their level of understanding and user experience. By collecting participants' preferences and feedback on the different visualization designs, the study analyzes the effectiveness of different visual variables and their combinations in representing speech prosody information, thereby evaluating the potential value of these visualization approaches in supporting pronunciation learning.

3 EVALUATION RESULTS

Table 2 shows that most participants preferred vertical text position for representing pitch and text size was perceived as the most intuitive feature for representing speech intensity.

Table 2: Users' visual mapping preference

Feature	Visual mapping	Percentage
Pitch	Text Size	6.70%
	Color	6.70%
	Vertical Position	86.70%
Intensity	Text Size	66.70%
	Color	26.70%
	Vertical Position	6.70%

Based on the preference results shown in the previous evaluation, a combined representation method was produced, shown as Figure 4, in which vertical text position represents pitch and text size represents speech intensity.

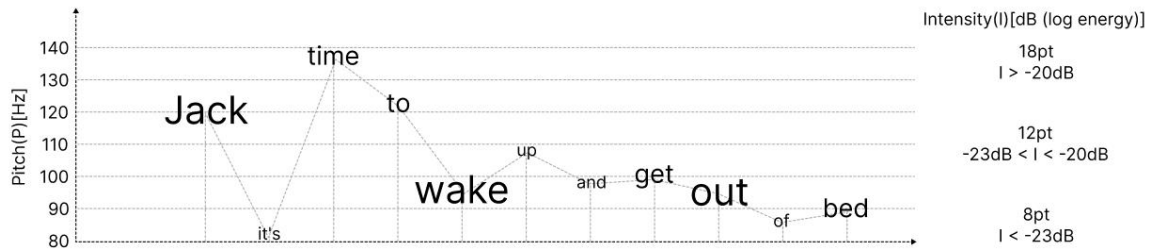


Figure 4: Fused Display of Prosodic Visual Mapping

In order to evaluate the effectiveness of this visualization method, multiple aspects of the system were rated using a 5-point Likert scale. The related results were analyzed by calculating the Mean (M) and Standard Deviation (SD) in Figure 5.

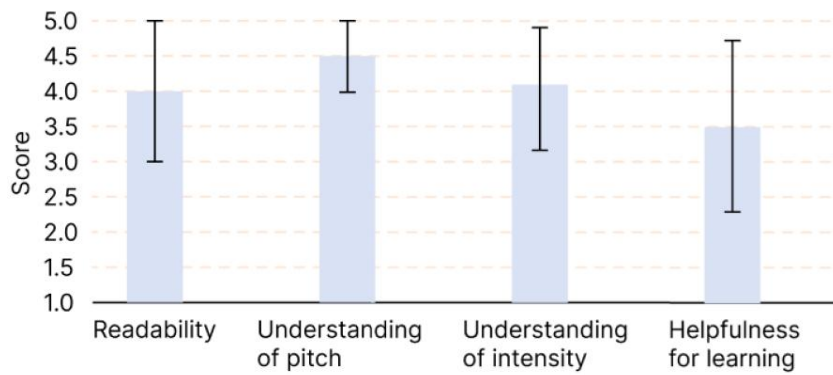


Figure 5: Mean user ratings ($\pm$ SD) of the proposed integrated prosody visual mapping on four evaluation criteria.

As shown in figure, indicating that the proposed visualization method can effectively help users perceive and interpret prosodic features in speech, and the visualization method can present speech prosody in a clear and intuitive way, thereby supporting its potential as an assistive tool for non-native pronunciation learning.

4 FUTURE WORK

From this study we gathered that we will focus on exploring the possibility of integrating this visualization method with gamified learning environments. One potential direction is to combine this visualization technology with mechanisms derived from 2D platform games, such as Mario-style jumping adventure games.

This gamified interaction approach can serve as a guided learning tool. Gamification can create a more engaging and interactive way to deliver learning material, integrates game design elements and principles into educational activities to enhance engagement, motivation, and learning outcomes, enriching the learning experience for students[6]. By embedding prosody training into interactive gameplay, it can help learners improve their perception and production of speech prosody in a more engaging and intuitive way.

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